

GAIT

Gait refers to a pattern or sequence of movements followed by a robotic system to move in space. The choice of gait depends on a variety of factors such as type of robotic system, number of legs, type of terrain that the robot is expected to navigate, speed, stability, energy and efficiency requirements. The chosen gait can be modified by varying the gait parameters such as duty factor, leg phase, leg stroke and cycle time.

Definition of Gait

A gait is defined by the time and the location of the placing and lifting of each foot, coordinated with the motion of the body in its six degrees of freedom, in order to move the body from one place to another.

DUTY FACTOR

Duty Factor (D) is a factor that describes what part of a walking cycle a leg spends in stance phase , i.e. the leg is in contact with ground in a support position.

$$D = \frac{T_s}{T_c}$$

Where

T_s : Time period leg is in stance phase

T_c : Cycle time of walking motion

For trot gait alternate pair of diagonal legs execute motion periodically. We pair our legs as Pair 1 (LF and RH) and Pair 2 (RF and LH). The time period of motion to start and end for stance and swing phase of each pair depends on the duty factor.

Motion when $D = 0.5$

When duty factor is equal to 0.5, half of the cycle time is used by Pair 1 to execute swing and Pair 2 to execute stance. After half of the cycle time this pattern is reversed as seen in the diagram below.

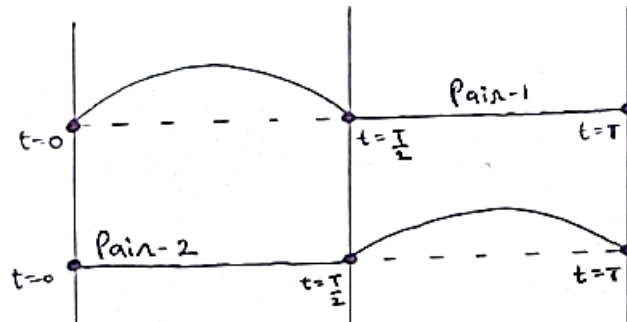


Figure: Representation of motion when Duty factor is 0.5

There is no overlap in swing or stance phase of pair of legs when duty factor is 0.5.

Motion when $D < 0.5$

When duty factor is less than 0.5, then more than half of the cycle time is used by Pair 1 to execute swing and Pair 2 spends less than half of the cycle time in stance.

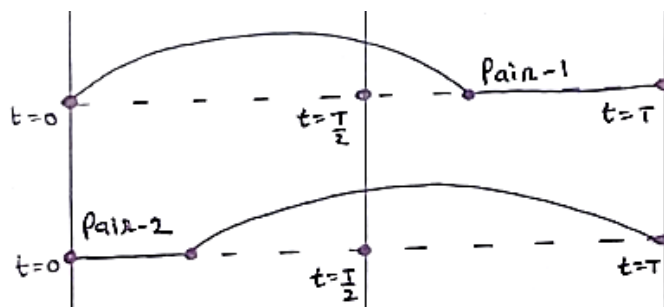


Figure: Representation of motion when Duty factor is less than 0.5

Once Pair 1 finishes swing it goes to stance and pair 2 starts swing once its stance phase ends. There is an overlap in swing phase of both the pairs.

Motion when $D > 0.5$

When duty factor is more than 0.5, then less than half of the cycle time is used by Pair 1 to execute swing and Pair 2 spends more than half of the cycle time in stance.

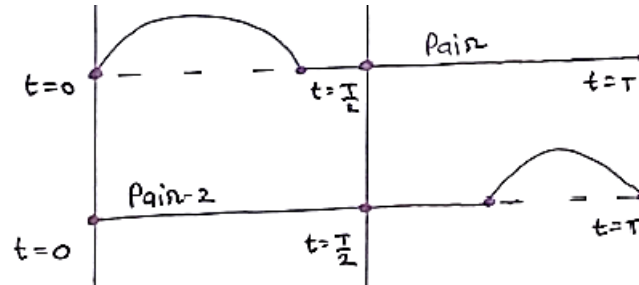


Figure: Representation of motion when Duty factor is more than 0.5

Once Pair 1 finishes swing it goes to stance and pair 2 starts swing once its stance phase ends. There is an overlap in stance phase of both the pairs.

TROT GAIT METHODOLOGY

To generate trot gait of desired duty factor we start with generation of required trajectory. We have chosen a quintic trajectory. Once the trajectory is chosen and generated, motion of the robot is initiated from its nominal position to bring the robot to a position ready to execute multiple walking cycles. Then time period for swing period and stance period are defined for each pair. Then starting one pair in swing and other in stance walking cycle is executed.

A difference in representation of stance motion of a pair when generating trajectory and gait should be noted when motion is shown w.r.t. torso as reference point and when an universal reference point is used. When trajectory of a foot and gait pattern is generated w.r.t. torso we show the motion of the foot from end position to start position to show the motion of the body, this is done as the torso (as reference point) remains stationary and when seen from torso it appears that the foot is retracing its path along the ground. In case of gait and trajectory generation when an universal reference frame is chosen, motion of torso and legs are observed individually.

Terminology used

S: Stride length

L: Step length

$$S = 2L$$

Frequency: Number of gait cycles completed per unit time

T: Time period of walking cycle

V: Velocity

$$V = S/T$$

D: Duty factor

$$\text{Stance duration} = DT$$

$$\text{Swing duration} = (1-D)T$$

X_i : Initial X position of torso

Y_i : Initial Y position of torso

Z_i : Initial Z position of torso

Sample input values

The following sample values are used as input to get output plots

$$S=0.1 \text{ m}$$

$$T=0.5 \text{ s}$$

$$X_i : 0$$

$$Y_i : 0$$

$$Z_i : 0$$

$$D = 0.5 \text{ (for balanced trot gait)}$$

$$D = 0.6 \text{ (for walking trot gait)}$$

$$V_x = 1 \text{ m/s}$$

$$V_z = 0.1 \text{ m/s}$$

$$H = 0.05 \text{ m}$$

TROT GAIT GENERATION

We are going to generate trot gait for three cases. When duty factor is equal to 0.5 we will generate 'Balanced trot gait', when duty factor is less than 0.5 we will generate 'Running trot gait' and when duty factor is more than 0.5 we will generate 'Walking trot gait'.

Balanced trot gait

In balanced trot gait the duty factor is 0.5, this pattern gives equal emphasis to stability and efficiency. In this pattern we expect to see motion of pair 1 to be executed first, immediately followed by the next pair (pair 2) as seen in normal walking.

Single Cycle of Balanced trot gait

Single cycle of a balanced trot gait follows the pattern given below (shaded portion shows leg in support phase):

Leg	Balanced Trot Gait			
1				
2				
3				
4				
Time	T/2			

Figure: Timing diagram for Single cycle of Balanced Trot Gait

The output graph obtained is:

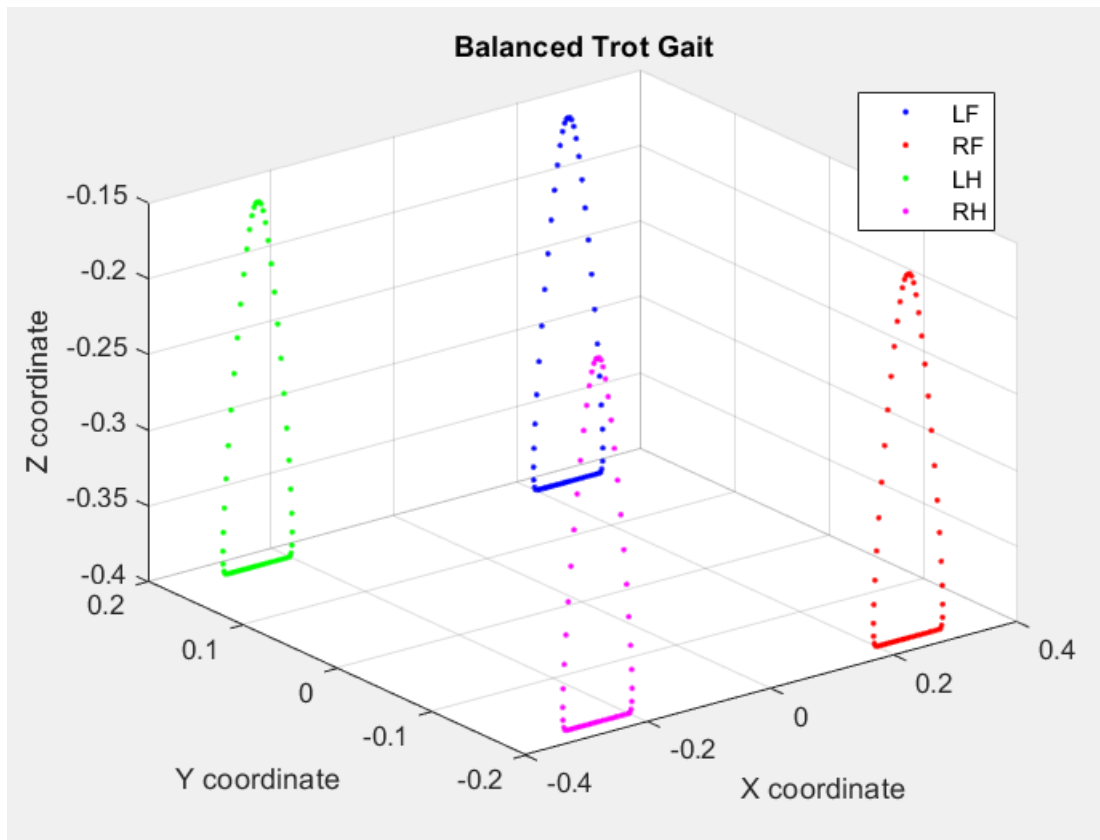


Figure: Output for single cycle of Balanced Trot Gait

To get a better idea of the motion we observe three consecutive walking cycles.

Multiple Cycles of Balanced trot gait

Multiple cycles of a balanced trot gait follows the pattern given below (shaded portion shows leg in support phase):

Leg	Balanced Trot Gait											
1												
2												
3												
4												
Time	$T/2$				$3T/2$				$5T/2$			

Figure: Timing diagram for Multiple cycles of Balanced Trot Gait

The output graph obtained is:

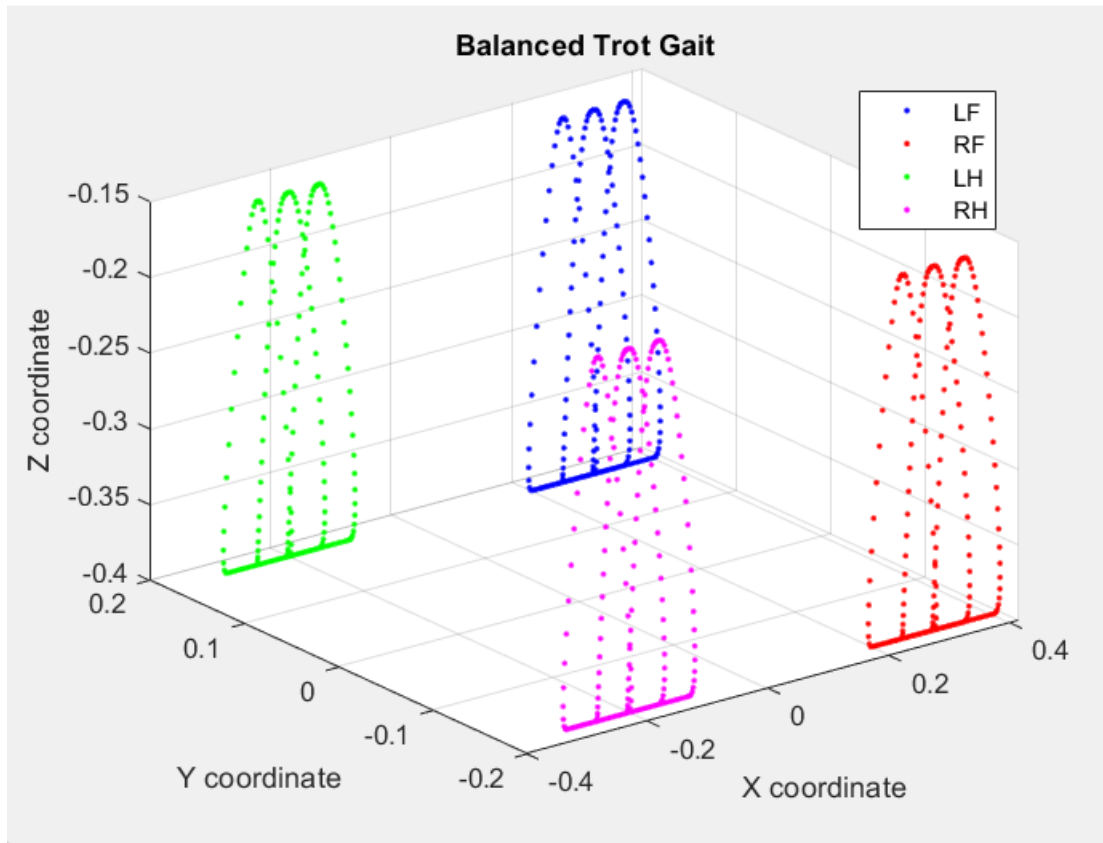


Figure: Output for multiple cycles of Balanced Trot Gait

Walking trot gait

In walking trot gait the duty factor is more than 0.5, this pattern gives more emphasis to stability compared to efficiency. In this pattern we expect to observe a gap or pause between the execution of swing of pair 1 and pair 2.

Single Cycle of Walking trot gait

Single cycle of a walking trot gait follows the pattern given below (shaded portion shows leg in support phase):

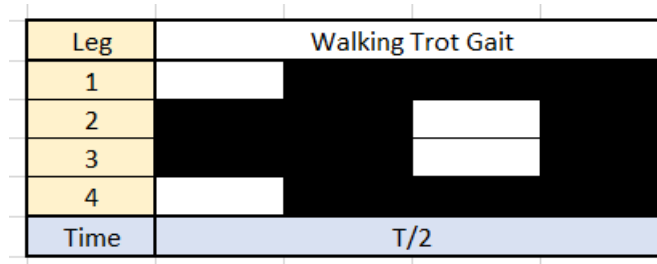


Figure: Timing diagram for single cycle Walking Trot Gait

The output graph obtained is:

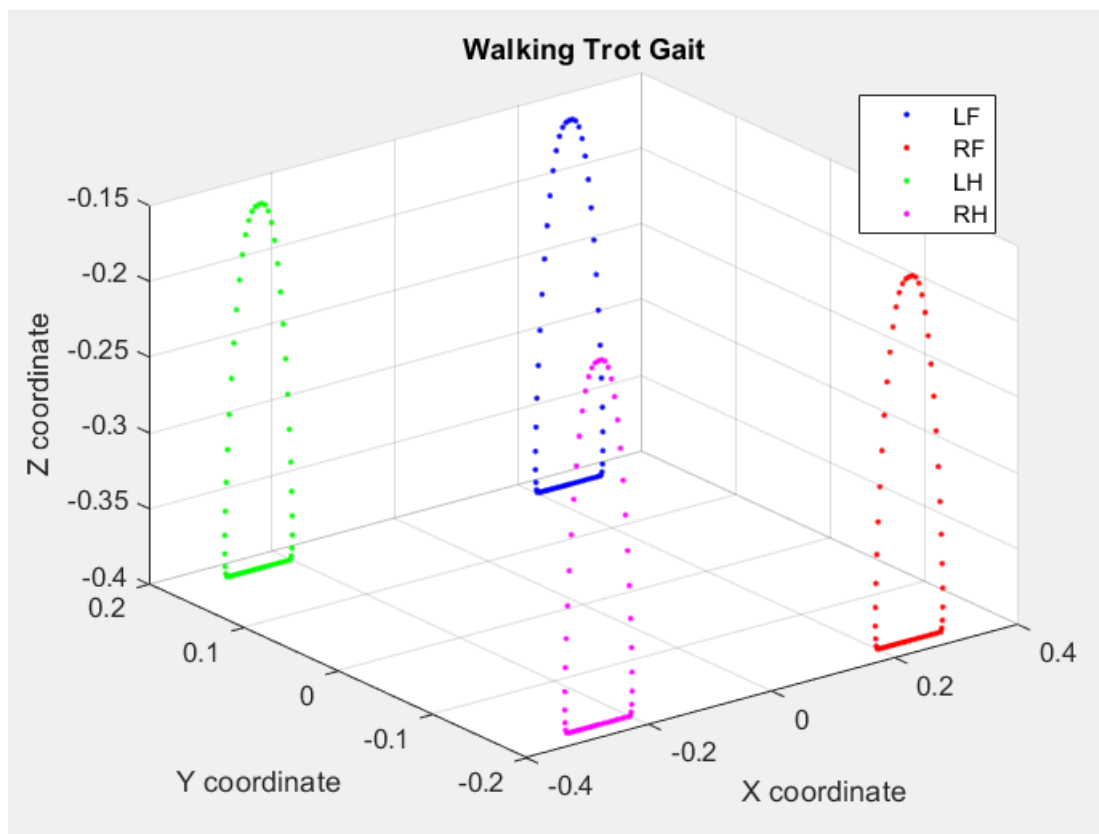


Figure: Output for single cycle Walking Trot Gait

To get a better idea of the motion we observe three consecutive walking cycles.

Multiple Cycles of Walking trot gait

Multiple cycles of a walking trot gait follows the pattern given below(shaded portion shows leg in support phase):

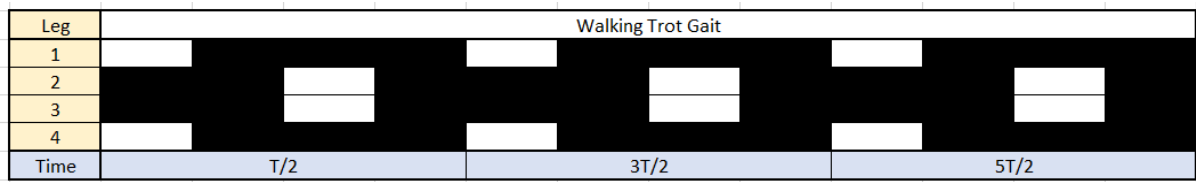


Figure: Timing diagram for Multiple cycles of Walking Trot Gait

The output graph obtained is:

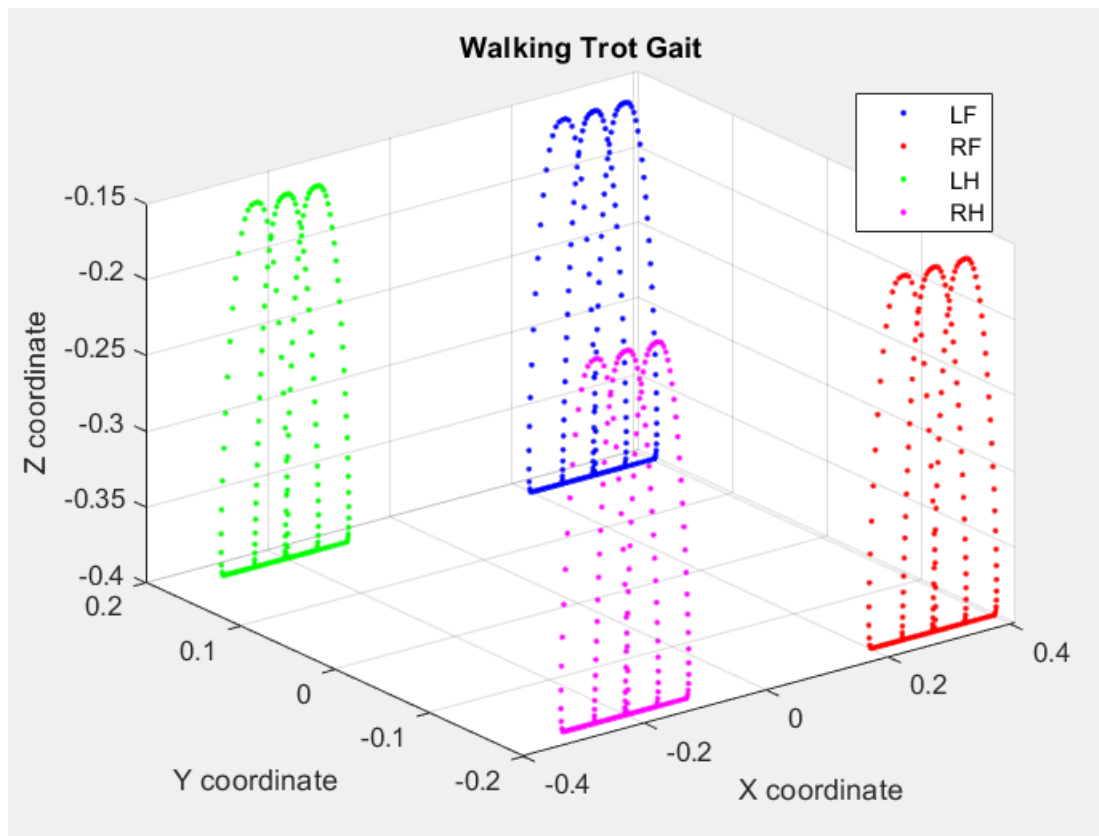


Figure: Output for Multiple cycles of Walking Trot Gait

Running trot gait

In running trot gait the duty factor is less than 0.5, this pattern gives more emphasis to speed of motion. In this pattern we observe to see that after pair 1 starts motion, pair 2 also starts motion before motion of pair 1 ends. Essentially there is an overlap of swing phase.

Physically this signifies that the robot executing this gait pattern will for a brief period during its gait cycle not have any support in form of contact with the ground. The robot will seem to make small jumps or gallops similar to that seen in horses when they run at a fast pace.

Single cycle of Running trot gait

Single cycle of a running trot gait follows the pattern given below (shaded portion shows leg in support phase):

Leg	Running Trot Gait			
1				
2				
3				
4				
Time	T/2			

Figure: Timing diagram for single cycle Running Trot Gait

Multiple cycle of Running trot gait

Multiple cycles of a running trot gait follows the pattern given below (shaded portion shows leg in support phase):

Leg	Running Trot Gait											
1												
2												
3												
4												
Time	T/2				3T/2				5T/2			

Figure: Timing diagram for multiple cycles of Running Trot Gait

WALKING CYCLES WITH DIFFERENT VELOCITIES

So far the gaits generated by us have the same initial swing speed for each cycle, that limits us to a particular walking cycle whenever we want to use that particular gait. It is desired that we should have the option of increasing or decreasing the speed at the start of each walking cycle as per our need. In this section we will attempt to change the speed of each walking cycle.

Methodology followed to generate walking cycles with different velocities

Gait generation is done in the same manner as discussed above using a quintic trajectory adopted for trot gait. We desire to change the speed for each walking cycle smoothly i.e. while maintaining stability and without any jerks. To maintain stability we have fixed the time period of each cycle as 0.4s, and to avoid jerks we have opted to change the velocity at the start of each walking cycle. Since the time period of motion is fixed change in walking speed is effected through change in stride length of that cycle. On increasing the velocity we expect to observe an increase in stride length and on decreasing the velocity we expect a decrease in stride length. To keep computations less expensive we have taken duty factor to be 0.5 and will plot motion for two walking cycles. The user is asked to give the desired starting velocities for each walking cycle.

We have initiated the motion by using a combination of movement of torso and legs pair 2. We will start our walking cycles with pair 1 in swing and pair 2 in stance and then keep alternating the state of motion as per the number of walking cycles. To smoothly implement these walking cycles we will move the body and legs of pair 2 to starting position from their nominal position, in interest of stability we will complete this motion in time equal to one walking cycle period i.e. 0.4s. First we calculate the stride length for first cycle based on user input, now for 0.2s the torso is first moved with a constant velocity for a distance equal to

half of the calculated, and then for the next 0.2s pair 2 executes complete motion. This is seen in the graph and may seem to be half of a walking cycle, but it is the initiation we have opted for smooth implementation of subsequent walking cycles.

Now after initiation we start with our walking cycles, during each walking cycle the torso is in continuous uniform motion and the diagonal pairs of legs (pair 1 and pair 2) alternatively execute motion. Stride length of each walking cycle varies and is calculated based on the velocity input by user. We expect to see different stride lengths in our walking graph. To study and analyse change in velocity we plot graphs of velocity of leg and torso w.r.t. time. We expect sudden spikes and periods of uniform motion for the plot of torso's speed, no spike is expected in speeds of initiation phase and first walking cycle. For the legs speed plot we expect to see no spike in maximum value of velocity between initiation and first walking cycle, between walking cycles a spike in maximum velocities of each cycles with smooth variation in between the period of spikes is expected.

Observations

To check whether the program written to implement trot gait with variable walking speeds for each walking cycle meets our expectations we test it. The sample values used were :

$$V_{\text{first cycle}} = 0.2 \text{ m/s}$$

$$V_{\text{second cycle}} = 0.8 \text{ m/s}$$

The output plot of the walking pattern obtained is:

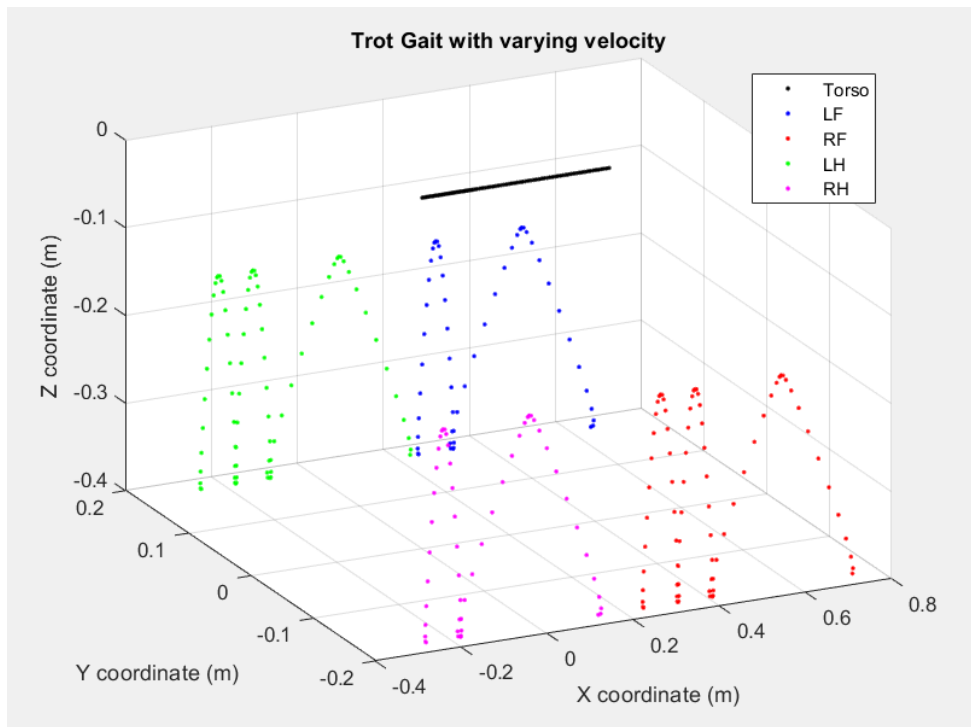


Figure: Walking Pattern of Trot Gait with variable velocity for each walking cycle

The output plot of the leg velocity w.r.t. time obtained is:

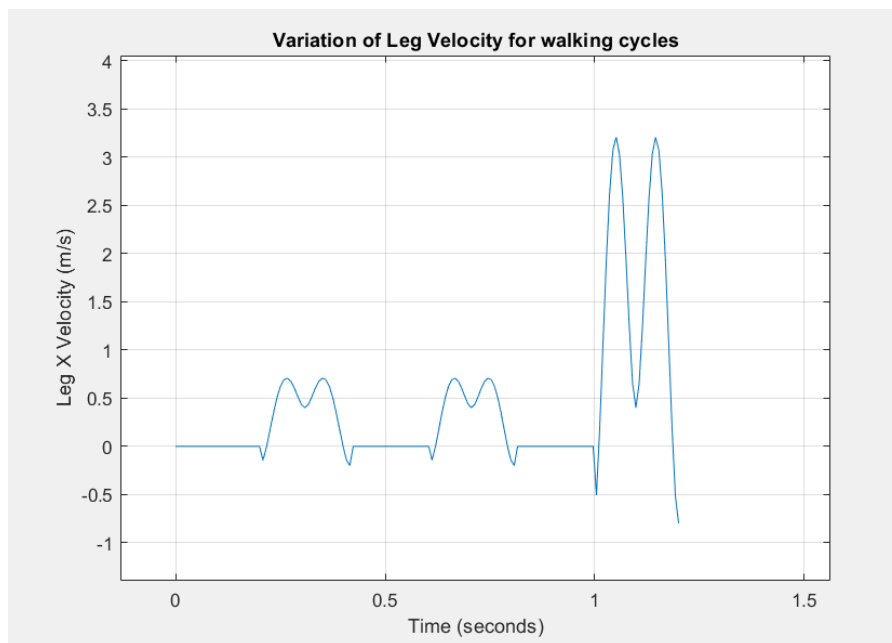


Figure: Variation of leg velocity w.r.t. time